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EE-102, Section 003

Lab 7 : Finite State Machine

Experiment 7

29.11.2023

AIM:

This lab project aimed to design a circuit as a Finite State Machine (FSM) on the breadboard. We will set up and observe results the state template we have prepared by LEDs and integrated circuit elements.

METHODOLOGY:

We wanted the situation template we prepared to be a real-life example. That's why we decided to have an FSM that shows a person's hunger and thirst state, and the change of these states is whether the person has money or not. First of all, we drew the state transition diagram to express the situations, and then we created the output table corresponding to that diagram. Then, we built the circuit corresponding to the result we obtained on the breadboard. Finally, we tested the situations by observing our inputs and outputs through the LEDs we connected.

DESIGN SPECIFICATIONS:

Our design has 1 input, 1 output and 1Hz clock input which are :

Table1 : In and out's of design

IN	Money	1 bit	Input that expresses whether a person has money or not
OUT	Human State	2 bit	Output showing the person's status
IN	clock	1 bit	1Hz clock from signal generator

And there are 4 states of FSM which are:

Table2 : Human State

Human State	00	Person is hungry and thirsty
	01	Person is thirsty but not hungry
	10	Person is not thirtsty but hungry
	11	Person is not thirsty and not hungry

At first, the person is hungry and thirsty("00"). As long as a person does not have money, he will remain hungry and thirsty in his initial situation. If he/she has money, his first choice will be to quench his thirst, which will lead him to the situation where he is not thirsty but hungry("10"). If he/she continues to have money in this state, it will satisfy his hunger and take him to the state of not being thirsty and not hungry ("11"), but if he/she does not have money, he/she will return to his initial state, that is, hungry and thirsty("00"). If the person continues to have money while being full and has water, he will maintain his situation("11"), but if he/she does not have money, he/she will first become thirsty but full("01"). If he/she has money while he is thirsty but full, he/she returns to the state where he is full and not thirsty("11"), but if he/she does not have money, he falls into his initial state of being hungry and thirsty. The state transistion diagram can be seen in Figure 1 and the output table can be seen in Table 3.

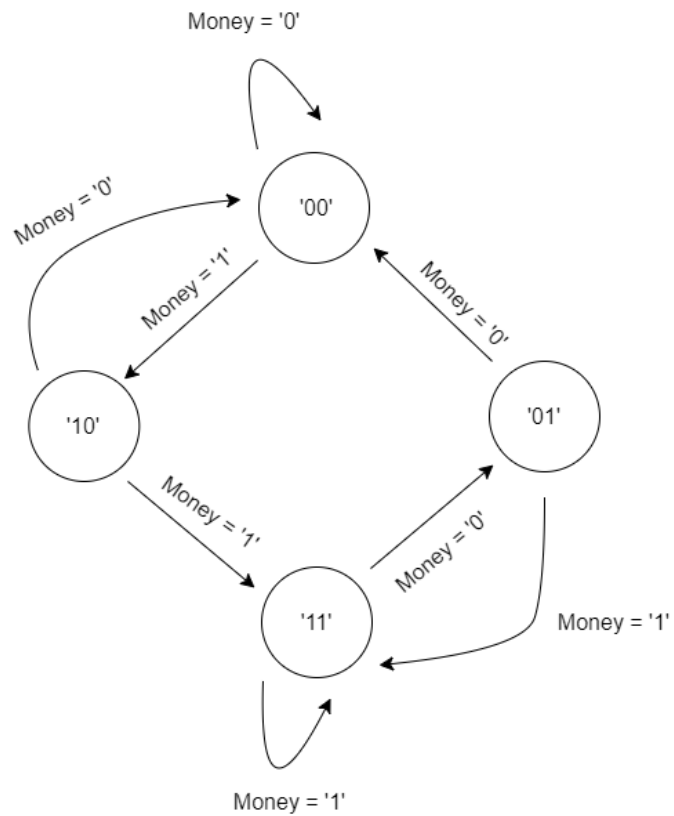


Figure 1 : State Transistion Diagram

Table 3 : Output Table

Current State		Input	Next State		Output
Q1	Q0	Money	Q1'	Q0'	Human State
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	1	1	1

Also, the schematic of design can be seen in Figure 2.

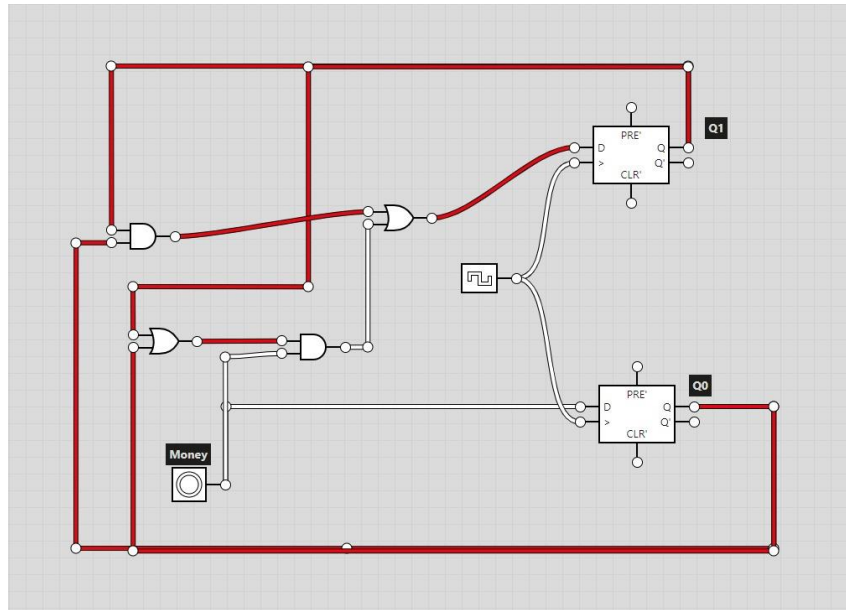


Figure 2 : Schematic of FSM

The design includes two D Flip Flops (two channels of 74HC74N), two OR gates (two channels of 74HC32N), two AND gates (one channel of 74HC08N).

RESULTS:

Implemented circuit on the breadboard can be seen in Figure 3.

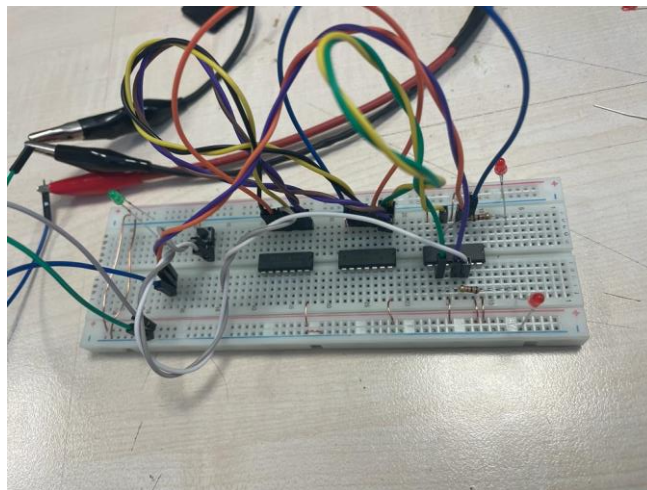


Figure 3: Implemented Circuit On The Breadboard

And some cases are tested:

CASE1:

Previous State	'00'	Hungry and Thirsty
Next State	'10'	Not Thirsty, Hungry
Money Input	'1'	Has Money

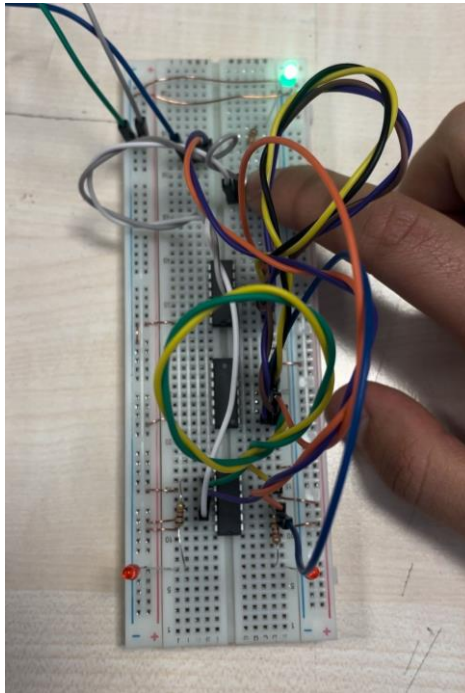


Figure 4 : Case 1 Previous State

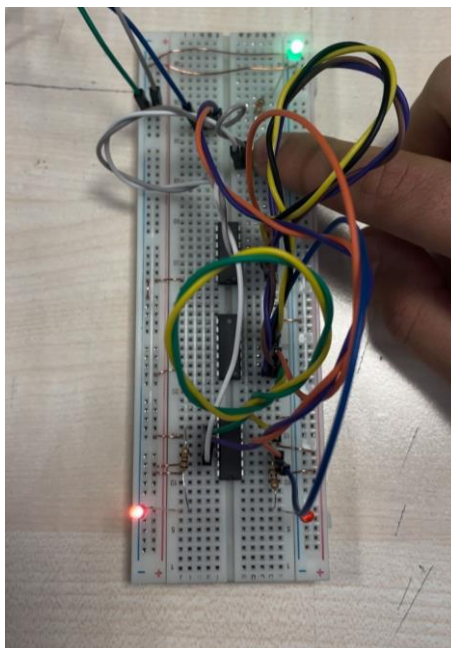


Figure 5: Case 1 Next State

CASE2:

Previous State	'10'	Not Thirsty, Hungry
Next State	'11'	Not Thirsty, Not Hungry
Money Input	'1'	Has Money

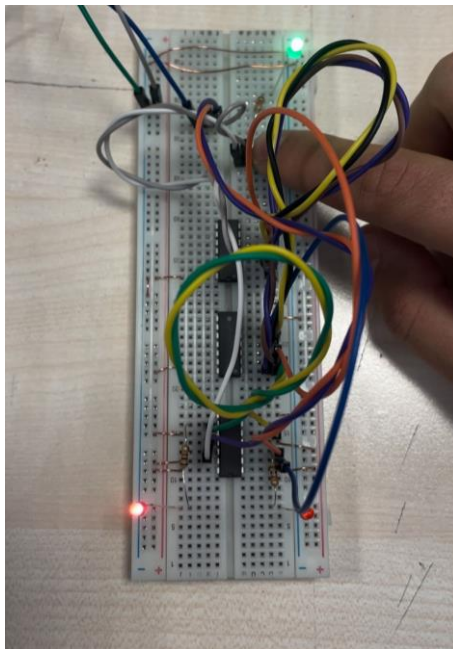


Figure 6 : Case 2 Previous State

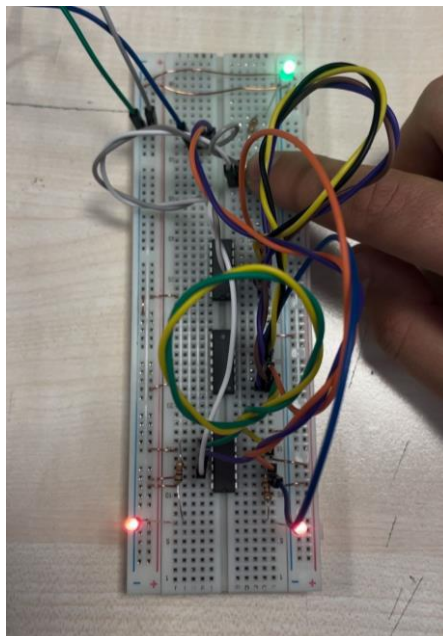


Figure 7 : Case 2 Next State

CASE3:

Previous State	'11'	Not Thirsty, Not Hungry
Next State	'01'	Not Hungry, Thirsty
Money Input	'0'	No Money

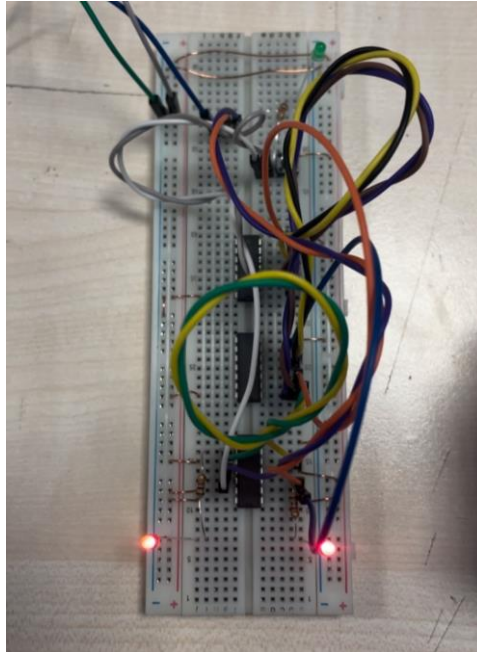


Figure 8 : Case 3 Previous State

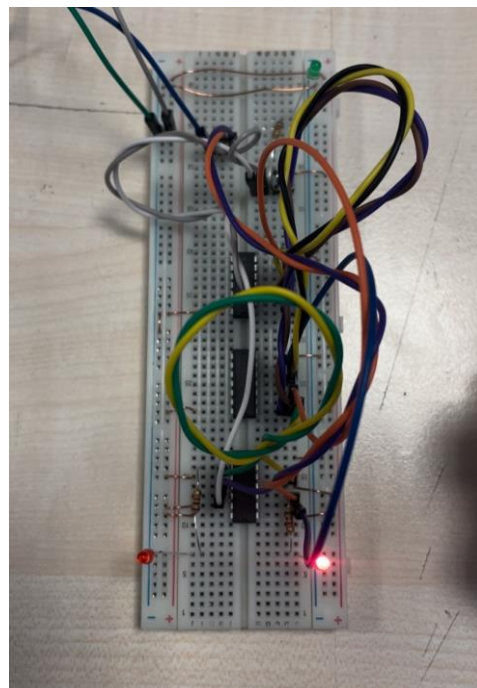


Figure 9 : Case 3 Next State

CASE4:

Previous State	'01'	Not Hungry, Thirsty
Next State	'00'	Hungry and Thirsty
Money Input	'0'	No Money

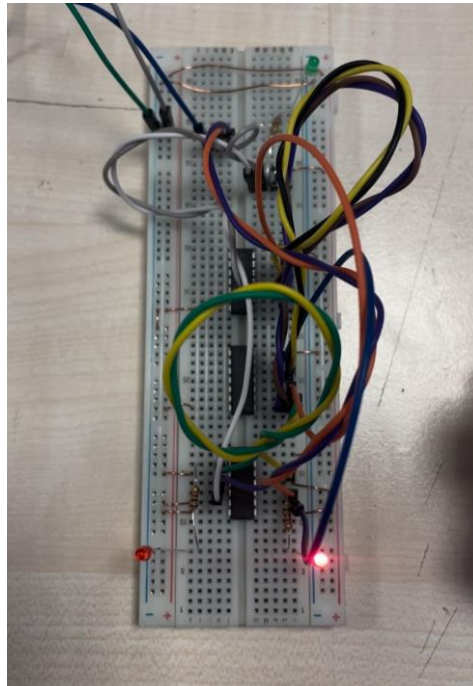


Figure 10 : Case 4 Previous State

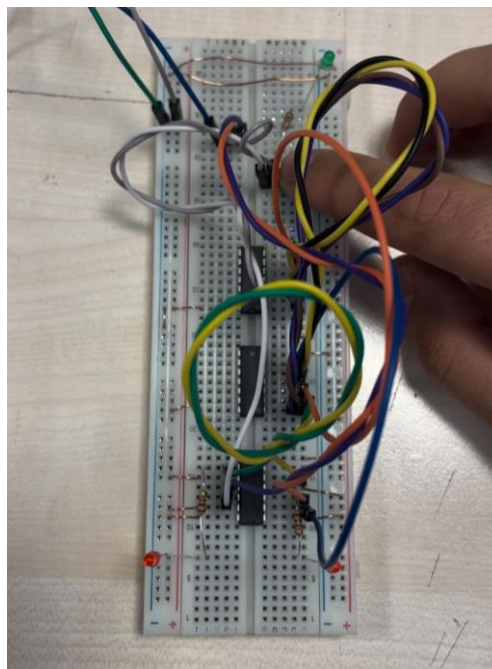


Figure 11 : Case 4 Next State

As can be seen that our design worked successfully and it reflects the state diagram to the experiment.

CONCLUSION:

In this laboratory, the aim was to design an Finite State Machine circuit and implement on a breadboard by integrated circuit elements. For this experiment, we preferred an FSM with one input and 4 states which reflects a real- life situation. We designed a situation where a hungry and thirsty person, having money, first quenches his thirst and then satisfies his hunger, and if he does not have money, he/she gradually becomes thirsty and hungry, then becomes hungry. We created a simple and understandable design by using the multiple inputs of integrated logic gates we have. The problem caused by the malfunction of a few integrated element was resolved by replacing the parts, and apart from that, no other problems were encountered. Our design worked as expected.